llmXive Automated Review of Agents' Last Exam

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and their supporting citations within the provided LaTeX source.

1. Discrepancy in Abstract Statistics The abstract states: "across mainstream harness and backbone configurations, the average full pass rate is 2.6%." However, Table 1 ('tables/main_results.tex') lists the "Overall Pass Rate" for all evaluated configurations. The lowest reported overall pass rate is 4.4% (Grok 4.3), and the highest is 26.2% (Codex/GPT-5.5). The 2.6% figure does not appear as a global average in the table. It is possible this figure refers specifically to the "Last-Exam" tier (where pass rates are 0.0% to 8.6%) or a specific subset of configurations not explicitly defined in the abstract sentence. The current phrasing suggests a global average across all configurations, which contradicts the data in Table 1. The authors should clarify whether this statistic applies to the "Last-Exam" tier specifically or if the calculation method differs from the table's "Overall Pass Rate."

2. Verification of External Benchmark Claims In Section 3.1 ("Main Results"), the text claims: "Codex... with GPT-5.5, the strongest configuration that achieves 82% on Terminal-Bench, reaches only a 25.2% overall pass rate on ALE-CLI." - The claim relies on the citation 'merrill2026terminalbench' for the 82% figure. Since the bibliography file ('references.bib') is not provided in the input, I cannot verify if the cited paper explicitly states "82%" for this specific model/harness combination. - The comparison implies a direct difficulty contrast. While the drop in performance is evident, the claim that ALE-CLI is "substantially harder" depends on the validity of the 82% baseline. If the 82% figure in the cited paper refers to a different subset of tasks or a different evaluation protocol (e.g., different time limits or task types), the comparison may be misleading. The text should ensure the citation supports the exact 82% figure and briefly note any protocol differences if they exist.

3. Consistency of "Last-Exam" Tier Claims The abstract mentions the "hardest tier remains far from saturated" with a 2.6% average pass rate. Section 3.1 defines the "Last-Exam" tier as comprising 35 tasks where "most agents achieve a 0% pass rate." Table 1 shows pass rates for the Last-Exam tier ranging from 0.0% to 8.6%. The 2.6% figure likely represents the average of these specific tier results. However, the abstract's phrasing "across mainstream harness and backbone configurations" is ambiguous. It should be explicitly linked to the "Last-Exam" tier to avoid confusion with the global averages in Table 1.

Conclusion The paper presents strong claims about benchmark performance and comparisons to existing benchmarks. While the internal data (Table 1) is consistent, the abstract's summary statistic (2.6%) appears to be a subset average presented as a global one, and the external benchmark comparison (82% on Terminal-Bench) requires verification against the cited source to ensure the specific value and context are accurate. These are primarily writing/clarity issues regarding the scope of claims rather than fundamental scientific flaws, but they affect the accuracy of the factual summary.

- [writing] Abstract claims 2.6% average pass rate across configurations, but Table 1 shows lowest overall rate is 4.4%. Clarify if 2.6% refers only to the Last-Exam tier.
- [writing] Section 3.1 cites 82% Terminal-Bench score for Codex/GPT-5.5. Verify ‘merrill2026terminalbench‘ explicitly reports this exact figure and configuration.
- [writing] Section 3.1 claims ALE-CLI is 'substantially harder' based on a drop from 82% to 25.2%. Ensure the 82% baseline is directly comparable in protocol and difficulty.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
Figure 1 effectively visualizes performance vs. resource trade-offs with clear axes and legends, but the caption's incomplete table reference and ambiguous token unit definition reduce reproducibility and precision.

Figure 2
The figure effectively visualizes the correlation between public subset and full pool pass rates, but the caption contains a grammatical typo and the rendered image lacks a legend to map the bubble colors to the specific taxonomy clusters.

Figure 3
Figure 3 visually compares the impact of varying harnesses versus varying models on pass rates. However, the caption's stated spreads (5.3–6.0pp) do not accurately match the visual ranges for the 'Vary Harness (Opus 4.7 fixed)' group, and the legend does not fully clarify the different fixed models used in the 'Vary Harness' groups.

Figure 4
The heatmap effectively visualizes per-task scores across models and instances, but the

caption contains a missing number placeholder and the x-axis labels are cluttered.

Figure 5

The heatmap effectively visualizes performance across tasks and models, but the caption contains a missing number placeholder, and the x-axis labels are too crowded to be easily legible.

Figure 6

The heatmap effectively visualizes performance across the Last-Exam tier, but the caption contains a missing number placeholder and the dense x-axis labels reduce readability.

Figure 7

The figure provides a qualitative overview of the software ecosystem but suffers from a missing figure number in the caption and ambiguous spatial grouping for the central 'Visual & Media Arts' domain.

Figure 8

The figure effectively visualizes the task construction pipeline with clear step labels and supporting UI screenshots, but the feedback loop logic and the specific function of the backend notification text are not explicitly defined in the diagram or caption.

Figure 9

The figure effectively visualizes the agent taxonomy but contains a potential scientific inaccuracy regarding CLI agent tool capabilities and has a minor legend definition gap.

Figure 10

The figure is incomplete as it only renders panel (a) despite the caption describing four distinct panels (a-d). Additionally, the x-axis lacks tick marks, hindering precise data interpretation.

Figure 11

Figure 11 is a clear, high-quality teaser diagram that effectively visualizes the broad taxonomy of professional tasks covered by the 'Agents' Last Exam' benchmark. The central wheel categorizes domains, while surrounding panels provide concrete examples of realistic workflows (e.g., Manufacturing, Game Development) linked to their respective sectors, fully supporting the caption's claim.

Figure 12

The figure is generally clear and well-structured, but the claim in the caption that 'All subdomains receive non-zero coverage' (referring to the orange QC category) is contradicted by the data shown for 'Maritime & Port Operations', which lacks an orange bar.

- [writing] Figure 1: The caption references 'Table' without a number, making it impossible to verify the harness-backbone configurations shown in the bubbles.
- [science] Figure 1: The legend defines bubble area as proportional to 'Total tokens' (100M–1000M), but the caption states 'bubble area is proportional to total token consumption' without specifying units or scale, creating ambiguity in interpretation.
- [writing] Figure 1: Panel (a) x-axis label 'Total API Cost (USD)' lacks currency symbol formatting consistency with scientific convention; consider '$' prefix or explicit note.
- [writing] Figure 2: The caption contains a typo 'Point size $$ total task instances per

cluster' where a verb (e.g., 'indicates') is missing between the subject and the definition.

- [writing] Figure 2: The legend is missing from the rendered image; the caption defines point size but does not define the color mapping for the taxonomy clusters shown in the plot.
- [science] Figure 3: The 'Vary Harness (GPT-5.5 fixed)' group contains 'Codex', 'OpenClaw', 'ALE-Claw', 'Cursor', and 'Droid'. However, the caption states the spread is 5.3–6.0pp, while the visual range for this group is ~19% to ~25% (a 6pp spread). The 'Vary Harness (Opus 4.7 fixed)' group shows 'Cursor', 'ALE-Claw', 'Claude Code' with a range of ~14.5% to ~19% (a 4.5pp spread). The caption claims 5.3-6.0pp for the latter, but the visual data suggests a smaller spread. The text '5.3pp' and '6.0pp' are pl
- [writing] Figure 3: The legend distinguishes 'Harness effect' (orange) and 'Model effect' (blue), but the x-axis labels are 'Vary Harness (GPT-5.5 fixed)', 'Vary Harness (Opus 4.7 fixed)', and 'Vary Model (OpenClaw fixed)'. The first two groups are both 'Vary Harness' but use different fixed models, which is not reflected in the legend. The legend should clarify that orange represents varying harnesses (with different fixed models) and blue represents varying models (with a fixed harness).
- [writing] Figure 4: The caption contains a placeholder error: 'Near-Term tier (task instances)' is missing the count (likely 59, matching the plot title).
- [writing] Figure 4: The x-axis labels (model/harness names) are rotated and densely packed, making them difficult to read without zooming.
- [writing] Figure 5: The caption contains a placeholder error, reading 'Full-Spectrum tier (task instances)' with a missing integer count for the number of instances.
- [writing] Figure 5: The x-axis labels (model/harness combinations) are rotated and densely packed, making them difficult to read without significant zooming.
- [writing] Figure 6: The caption contains a placeholder error: 'Last-Exam tier (task instances)' is missing the specific count of instances.
- [writing] Figure 6: The x-axis labels are rotated and densely packed, making model names (e.g., 'OpenClaw / Grok 4.3') difficult to read.
- [science] Figure 6: The y-axis lists task instance names but lacks the color-coded domain indicators shown in the legend, forcing reliance on text color which is hard to distinguish.
- [writing] Figure 7: The caption contains a broken cross-reference ('appear in Figure .') where the figure number is missing.
- [science] Figure 7: The 'Visual & Media Arts' label is positioned in a central overlap region, but the associated icons (e.g., Unity, Blender) are scattered across the center and right, making the domain boundary ambiguous.
- [writing] Figure 8: The caption describes a linear pipeline, but the diagram includes a feedback loop from 'QC Committee Review' back to 'Expert Task' (Submission/Editing)

without a legend or label explaining the iteration logic.

- [writing] Figure 8: The text 'Website Display' and 'Backend Trigger & Email Notified' are positioned ambiguously between the pipeline steps and the UI screenshots, lacking clear arrows or connectors to indicate their specific role in the workflow.
- [science] Figure 9: The 'Hands' (Tools) row for CLI-Agents is marked 'Full', but CLI agents typically lack GUI tool interaction; the diagram implies CLI agents have full tool capabilities which contradicts the standard definition of CLI vs GUI agents.
- [writing] Figure 9: The legend at the bottom right is incomplete; it defines 'Full', 'Limited', and 'N/A' symbols, but the 'Limited' symbol (half-filled circle) is not explicitly defined in the text, relying on visual inference.
- [fatal] Figure 10: The caption describes four panels (a, b, c, d), but the rendered image displays only a single panel (domain-level mean scores). The figure is incomplete.
- [science] Figure 10: The x-axis label 'Mean score (%)' is present, but the axis lacks tick marks and grid lines, making it difficult to accurately read the specific values for each domain.
- [science] Figure 12: The legend defines orange as 'Unverified' (expert submissions awaiting QC), but the caption states 'All subdomains receive non-zero coverage.' The 'Transport. & Safety' subdomain 'Maritime & Port Operations' shows only a blue bar (13) and no orange bar, contradicting the claim that all subdomains have unverified submissions.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on domain-specific acronyms and technical shorthand that obscure meaning for a general scientific audience. The most critical issue is the introduction and subsequent overuse of the acronym **GCUA** (Generalist Computer-Use Agent). While the full term is defined once, the paper immediately shifts to using "GCUA" as a standalone noun (e.g., "evaluate all agent systems in GCUA configuration," "GCUA reference"). This creates a barrier for readers not deeply embedded in the specific "computer-use agent" sub-field. The term should be replaced with "generalist agent" or the full phrase to improve readability.

Similarly, **CUA** (Computer-Use Agent) and **MCP** (Model Context Protocol) are used without explicit definition in the main text. "CUA" appears in the phrase "CUA MCP bridge" in Section 2.2 and Appendix A.1, assuming the reader knows both the agent type and the protocol. "MCP" is a specific industry standard that requires a brief explanation or citation upon first mention.

The term **harness** is used extensively to describe the agent's orchestration layer (e.g., "mainstream harness implementations," "harness sweeps"). While standard in software testing, in this context it functions as jargon for "agent framework" or "orchestration system.” Replacing ”harness” with ”framework” would make the text more accessible.

Additionally, **SOC** (Standard Occupational Classification) is referenced in the Abstract and Section 2.2 without being spelled out. While ”O*NET” is also an acronym, it is often more recognizable in this context; however, ”SOC” is a specific US government taxonomy that should be defined. Finally, **backbone** is used to refer to the underlying foundation model (e.g., ”fixed backbone,” ”backbone configurations”). This is deep learning jargon that should be replaced with ”foundation model” or ”base model” to align with broader AI literature and aid non-specialist readers.

These changes are purely stylistic and do not affect the scientific validity of the work, but they are necessary to ensure the paper is accessible to the broader AI community and interdisciplinary readers.

- [writing] Define ’CUA’ (Computer-Use Agent) at first use in Section 2.2. The text currently uses ’CUA’ and ’GCUA’ without explicitly spelling out the acronym ’Computer-Use Agent’ in the main body, relying on the reader to infer it from context or the appendix.
- [writing] Replace the acronym ’GCUA’ with ’Generalist Computer-Use Agent’ or ’generalist agent’ throughout the text. The term ’GCUA’ is introduced but then used repeatedly as a standalone noun (e.g., ’evaluate all agent systems in GCUA configuration’), which is non-standard and excludes non-specialist readers.
- [writing] Define ’MCP’ (Model Context Protocol) at first use in Section 2.2 and Appendix A.1. The text mentions ’CUA MCP bridge’ and ’MCP server’ without defining the protocol, assuming reader familiarity with a specific industry standard.
- [writing] Replace ’harness’ with ’orchestration framework’ or ’agent framework’ in the main text. While ’harness’ is common in testing, its use to describe the entire agent loop (e.g., ’mainstream harness implementations’) is jargon-heavy and less accessible than ’framework’ or ’system’.
- [writing] Define ’SOC’ (Standard Occupational Classification) at first use in the Abstract and Section 2.2. The text references ’SOC 2018’ immediately without spelling out the classification system, which is not universally known outside labor economics or specific US government contexts.
- [writing] Replace ’backbone’ with ’foundation model’ or ’base model’ in the main text. The term ’backbone’ (e.g., ’backbone configurations’, ’fixed backbone’) is technical jargon from deep learning architecture that may confuse readers from other domains.
- [writing] Define ’QC’ (Quality Control) at first use in Figure 2 caption. The caption mentions ’Quality Control (QC) Process’ but the acronym ’QC’ is used later in the text without re-definition or context for non-industry readers.

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a logically sound framework for the Agents' Last Exam (ALE) benchmark, with clear definitions of the agent architecture and evaluation pipeline. However, there are three specific areas where the text's claims or descriptions create logical friction with the provided data or standard interpretations of the benchmark's design.

First, in Section 2.3 ("Task Construction Pipeline"), the authors state that the public set includes the "full Last-Exam tier" while the private pool contains "proportionally more Near-Term-level tasks." Given that "Last-Exam" is defined in Section 3.1 as the hardest tier and "Near-Term" as the easiest, this phrasing implies the private pool is, on average, easier than the public set. This contradicts the typical logic of benchmark design, where the most difficult or "gold" tasks are often kept private to prevent overfitting, or where the public set is a representative sample. If the private pool is indeed easier, the claim that the public set is "representative" (verified in Appendix A.1) needs stronger justification, as a public set skewed toward the hardest tasks would not be representative of the full pool's average difficulty. The authors should clarify the distribution logic to resolve this apparent contradiction.

Second, in Appendix A.4 ("Failure Taxonomy"), the text explicitly states that "Timeout and resource-exhaustion cases are excluded from this breakdown" because they reflect environment constraints. However, Table A.3 ("Timeout frequency by difficulty tier") prominently displays "Timeout rate" and "Timeout score" as key metrics. While the exclusion from the *taxonomy pie chart* (Figure 4d) is logical, the text's phrasing "excluded from this breakdown" is ambiguous. It risks leading the reader to believe these cases are ignored entirely in the analysis. The text should explicitly state that timeouts are excluded from the *root-cause taxonomy* but are fully accounted for in the *performance statistics* to ensure the reader understands the distinction between the two analyses.

Third, the Abstract and Section 3.1 claim that "across mainstream harness and backbone configurations, the average full pass rate is 2.6%" for the hardest tier. A review of Table 1 reveals that the top configurations (e.g., Codex with GPT-5.5) achieve an 8.6% pass rate on the Last-Exam tier, and even mid-tier models often exceed 2-3%. The figure of 2.6% appears inconsistent with the data presented in the table unless it refers to a specific, undefined subset of configurations or a different calculation method. To maintain logical consistency, the authors should either correct the statistic to match the table data or explicitly define the subset of models used to calculate the 2.6% average.

- [writing] Section 2.3 claims the public set includes the 'full Last-Exam tier' while the private pool has 'more Near-Term tasks.' This implies the private pool is easier, contradicting standard benchmark design where hard tasks are often kept private. Clarify the distribution logic to resolve this apparent contradiction with the 'Last-Exam' difficulty definition.
- [writing] Appendix A.4 states timeout cases are 'excluded from this breakdown' (the taxonomy), yet Table A.3 explicitly reports timeout rates and scores. Clarify that timeouts

are excluded from the *taxonomy analysis* but included in *performance metrics* to avoid confusion about whether they are counted as failures.

- [science] The Abstract and Section 3.1 cite an 'average full pass rate is 2.6%' for the hardest tier. Table 1 shows top models achieving 8.6% on Last-Exam. Verify this statistic against the table data or specify the exact subset of configurations used to derive 2.6% to ensure consistency.

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several strong claims regarding the scope and impact of the Agents' Last Exam (ALE) benchmark that extend beyond the immediate evidence presented in the manuscript.

First, the claim in the Related Work section (Table 1) and Introduction that ALE is the "first benchmark to cover all 55 SOC/O*NET industries" is an overstatement of the taxonomy's alignment with the federal standard. Appendix A.1 ("Taxonomy Details") explicitly clarifies that the final taxonomy consists of 51 subdomains anchored in SOC 2018/O*NET, supplemented by "four frontier subdomains" derived from recent roadmaps and not present in the official 2018 classification. By claiming coverage of "all" SOC industries, the authors conflate their extended taxonomy with the official federal standard. This should be corrected to state that ALE covers the *extended* taxonomy or the specific set of 55 subdomains defined in the paper, rather than implying complete coverage of the SOC 2018 standard itself.

Second, the abstract and introduction frame ALE as an instrument "intended not merely as another leaderboard, but as an instrument for closing the gap between benchmark success and GDP relevant impact." While this is a compelling vision, the paper provides no empirical data linking ALE scores to actual economic output, GDP contribution, or real-world deployment success. The results show that current agents score low (2.6% full pass rate), but this does not prove that passing ALE would result in economic transformation. This framing overreaches the scientific evidence; the text should be tempered to reflect that ALE is a *proxy* or *candidate instrument* for measuring such impact, rather than a proven mechanism for closing the gap.

Finally, the assertion in Section 3.1 that "ALE-CLI tasks are substantially harder" than Terminal-Bench relies primarily on a single performance comparison (Codex/GPT-5.5 achieving 25.2% on ALE-CLI vs. 82% on Terminal-Bench). While the performance gap is significant, the qualitative leap to "substantially harder" without a controlled comparison of task difficulty distributions (e.g., step counts, tool complexity, or domain specificity) risks over-interpreting the results. The text should qualify this as "observed lower performance" or "higher difficulty in the current evaluation" rather than an intrinsic hardness claim without further ablation or difficulty analysis.

- [writing] Claiming ALE covers 'all 55 SOC/O*NET industries' (Table 1) overreaches;

Appendix A.1 notes 4 subdomains are ’frontier extensions’ not in SOC 2018. Clarify that coverage applies to the paper’s extended taxonomy, not the federal standard itself.

- [writing] The claim that ALE will ’close the gap’ to GDP impact (Abstract/Intro) lacks empirical support linking scores to economic output. Temper this to state ALE is a ’proxy instrument intended’ to measure such impact, not a proven mechanism.
- [writing] Stating ALE-CLI tasks are ’substantially harder’ than Terminal-Bench (Sec 3.1) relies on a single performance gap without difficulty distribution analysis. Qualify as ’observed lower performance’ rather than an intrinsic hardness claim.

paper_reviewer_safety_ethics

Verdict: minor_revision
The paper presents a comprehensive benchmark for evaluating AI agents on professional workflows. From a safety and ethics perspective, the scope of the benchmark—spanning radiology, cybersecurity, and industrial control systems—introduces significant dual-use and privacy risks that require explicit mitigation strategies in the text.

First, regarding **human subjects and data privacy**, the benchmark relies on workflows sourced from 250+ industry experts (Section 3.3, Appendix B). While the paper mentions a ”submission portal,” it does not state whether the collection of these professional tasks and the associated data (which may include proprietary code, financial records, or patient data in the radiology examples) was conducted under an Institutional Review Board (IRB) protocol or with informed consent. Given the potential for these tasks to reveal sensitive professional practices or personal data, a formal ethics statement or a clear justification for exemption is necessary.

Second, the **dual-use potential** of the benchmark is high. The inclusion of tasks in cybersecurity (e.g., ‘snake_crackme‘) and the ability of agents to execute shell commands and interact with GUIs in industrial software (e.g., PowerMill, Moldex3D) creates a risk that the benchmark itself could be used to train agents for malicious purposes, such as automating cyberattacks or industrial sabotage. The paper currently describes the evaluation pipeline but lacks a dedicated section on **safety guardrails and containment**. Specifically, it should detail how the virtual machine environments are sandboxed to prevent agents from escaping to the host, accessing external networks, or modifying systems outside the task scope.

Third, specific **domain-sensitive tasks** require clarification. The radiology task (‘radiology/microdicom_nih_cxr_reader_adjudication‘) involves medical imaging. The authors must explicitly confirm that all patient data has been de-identified in accordance with HIPAA or GDPR standards and that no Protected Health Information (PHI) is present in the benchmark artifacts. Similarly, for cybersecurity tasks, the paper should clarify that the ”crackme” challenges are synthetic or derived from public, non-malicious sources, and that the benchmark does not facilitate the generation of real-world exploits.

Finally, the use of **LLM-as-judge** for certain tasks (Section 4.3, Appendix A.3) introduces risks of bias or hallucination in scoring, particularly for subjective or safety-critical domains. While the paper notes a preference for deterministic scoring, the 6.8% of tasks relying on LLM judges should be reviewed for potential safety failures (e.g., an agent generating harmful content that the judge fails to penalize). A brief discussion on the safety of the judging prompts and the potential for reward hacking in these specific cases would strengthen the ethical rigor of the evaluation.

In summary, while the benchmark is technically sound, the manuscript requires additions to address IRB/ethics compliance, dual-use risks, data privacy in sensitive domains, and containment strategies to meet safety and ethics standards.

- [writing] The manuscript lacks an explicit statement regarding IRB approval or ethics committee oversight for collecting workflows from 250– industry experts. Add a formal ethics statement or exemption justification in Section 3.3 or Appendix B.
- [writing] The 'GUI-as-Tool' mode grants agents full desktop control. The paper must explicitly detail sandboxing measures beyond standard VM isolation to prevent host escape, external network access, or modification of systems outside the task scope.
- [writing] Tasks in radiology and cybersecurity raise dual-use and privacy risks. Clarify HIPAA/GDPR compliance for patient data in radiology tasks and confirm cybersecurity tasks use synthetic/public data to prevent generating real exploits.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The paper presents a large-scale benchmark (ALE) with 150 public task instances and extensive experimental results. The scientific evidence regarding the benchmark's construction and the magnitude of current agent failures is generally strong, supported by a clear taxonomy and deterministic evaluation modes. However, there are specific gaps in the statistical rigor and validation of the analytical claims that require attention.

First, the claim that the public subset of tasks is representative of the full private pool (Appendix~\ref{app:fullpool}) is supported by a scatter plot showing a Pearson correlation of $r=0.89$ between pass rates on the public and full sets. Crucially, this analysis is performed for only **one** agent configuration (Claude Code – Opus 4.7). It is not demonstrated whether this correlation holds for other models or harnesses, which is a significant limitation given the paper's goal of providing a general evaluation instrument. The authors should either extend this analysis to multiple configurations or explicitly qualify the claim as specific to the tested configuration.

Second, the failure taxonomy analysis (Section~\ref{sec:experiment-analysis}, Appendix~\ref{app:failure-taxonomy}) relies entirely on an LLM (OpenAI Codex) to classify the root causes of agent failures into categories like "Understanding," "Approach," and "Execution." The manuscript does not provide any measure of inter-rater reliability (e.g., Cohen's kappa) or a human validation study to verify the accuracy of these automated classifications. Given that the paper draws major conclusions about the nature of current agent limitations (e.g., ”dominant bottleneck is domain knowledge”) from this taxonomy, the lack of validation for the classification method weakens the evidentiary support for these claims.

Finally, while Table~\ref{tab:main-results} reports standard deviations for a subset of runs, these appear to be derived from repeating the same task instance a few times rather than capturing the variance across the diverse set of 150 tasks. For a benchmark with such heterogeneous tasks, the standard error of the mean across tasks or the distribution of scores per task is essential to determine if the observed differences between harnesses (e.g., the 1-2% gaps) are statistically significant or within the noise of task difficulty variance. The current reporting makes it difficult to assess the robustness of the comparative results.

- [science] The claim that the public subset is representative of the full pool relies on a single correlation (r=0.89) for one agent configuration (Appendix~ ef{app:fullpool}). Validate this across multiple models or explicitly discuss the limitation.
- [science] The failure taxonomy relies on an LLM classifier without reported inter-rater reliability or human validation (Appendix~ ef{app:failure-taxonomy}). Add reliability metrics to support the conclusion that ’Understanding’ errors dominate.
- [science] Table~ ef{tab:main-results} reports standard deviations from repeated runs of single instances but lacks variance across the 150 diverse tasks. Report standard error of the mean across tasks to assess statistical significance of harness differences.

paper_reviewer_statistical_analysis

Verdict: minor_revision
The statistical reporting in the manuscript is generally clear but lacks specific details required for full reproducibility and rigorous interpretation of uncertainty.

First, regarding the error bars in **Table 1** (lines 1-15 of ‘tables/main_results.tex‘), the caption notes that ”Superscript ± values denote score standard deviations estimated from three independent runs.” However, it is ambiguous whether these standard deviations are calculated across the three runs of a *single* task instance (and then averaged) or if the three runs were aggregated into a single mean score per configuration and the SD reflects variance across tasks. Given the high variance in agent performance across different task types, the unit of analysis for the standard deviation is critical. The authors should explicitly state the aggregation hierarchy (e.g., ”SD of the mean score across 3 runs per task, averaged over N tasks”) to allow readers to assess the precision of the reported means.

Second, in **Appendix:fullpool-evaluation** (lines 1-15 of ‘appendix/fullpool-evaluation.tex‘), the authors report a Pearson correlation of $r=0.89$ with $p<0.001$ between public and full-pool pass rates across taxonomy clusters. The analysis is based on $n=13$ clusters (the number of top-level domains). With such a small sample size, the point estimate of the correlation is highly sensitive to outliers, and the p-value, while significant, does not convey the uncertainty of the estimate. The authors should report the 95% confidence interval for the correlation coefficient (e.g., via bootstrapping or Fisher’s z-transformation) to provide a more honest assessment of the representativeness claim.

Finally, the **failure taxonomy** analysis in **Appendix:failure-taxonomy-details** (lines 1-15 of ‘appendix/failure-taxonomy-details.tex‘) presents a distribution of failure modes (47% Approach, 31% Understanding, 22% Execution). The text notes that ”Timeout and resource-exhaustion cases are excluded from this breakdown.” To prevent misinterpretation of the prevalence of these failure modes, the authors should explicitly state the total number of failed runs analyzed and the proportion of total failures that were excluded due to timeouts. If timeouts constitute a significant portion of failures (e.g., >20%), the reported distribution of ”reasoning” failures may be biased toward the subset of tasks that agents actually attempted to solve rather than the full difficulty spectrum.

- [writing] Table 1 reports standard deviations (±) for a subset of configurations based on ’three independent runs’ but does not specify the unit of aggregation (e.g., mean of 3 runs per task vs. 3 runs per task instance). Clarify the statistical unit and aggregation method to ensure reproducibility of the error bars.
- [science] The paper claims a ’strong correlation’ (r=0.89, p<0.001) in Appendix Fig. 1 regarding public vs. full-pool representativeness. However, the sample size (n=13 clusters) is small for a robust Pearson correlation. Report the 95% confidence interval for the correlation coefficient and consider discussing the stability of this estimate given the low degrees of freedom.
- [writing] In the failure taxonomy analysis (Appendix), percentages sum to 100% of ’classifiable failures’ (47% + 31% + 22%). Explicitly state the total number of failed runs analyzed and the percentage of total failures that were excluded (e.g., due to timeout or infrastructure issues) to avoid selection bias in the reported distribution.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript demonstrates a high level of technical detail and generally clear prose, effectively communicating the scope and methodology of the Agents’ Last Exam (ALE) benchmark. The structure is logical, and the use of figures and tables to support the text is appropriate. However, there are a few areas where the writing could be refined to improve flow, clarity, and adherence to standard academic conventions.

First, in Section 2.1, the use of inline ‘\colorbox‘ commands to highlight ”Undesired” and ”Better” examples is unconventional for a research paper and disrupts the visual flow of the text. While the intent is to draw attention to the contrast, a standard table or a more integrated textual description would be more professional and easier to read.

Second, there are minor grammatical issues that affect clarity. For instance, in Section 3.2, the phrase ”The rare appearance single ’task’ refers to...” is awkward and likely a typo. It should be rephrased for grammatical correctness. Similarly, in Appendix A.1, the use of LaTeX-specific number formatting (e.g., ‘1{,}016‘) should be checked to ensure it renders correctly in the final PDF, as it may appear as ‘1,016‘ or ‘1 016‘ depending on the compiler, potentially causing confusion.

Finally, some sentences in the appendices, particularly in Appendix A.4, are slightly verbose. Tightening these sentences would improve the overall conciseness and readability of the paper without losing any essential information.

Overall, the writing is strong, but these minor revisions would elevate the manuscript to a higher standard of academic writing.

- [writing] In Section 2.1 (Benchmark Design Principles), the examples contrasting 'Undesired' and 'Better' tasks use inline color commands (\colorbox) that are visually distracting and non-standard for academic prose. Rephrase these as standard text or use a formal table to improve readability and flow.
- [writing] In Section 3.2 (Agent Architecture), the sentence 'The rare appearance single 'task' refers to the runnable instance level' contains a grammatical error and is unclear. It should be revised to 'The rare appearance of the single word 'task' refers to the runnable instance level' or similar for clarity.
- [writing] In Appendix A.1 (Taxonomy Definition), the phrase '1{,}016 entries' uses a LaTeX-specific number formatting command that may not render correctly in all viewers. Ensure consistent number formatting (e.g., 1,016) throughout the text for better readability.
- [writing] In Appendix A.4 (Task Construction Pipeline), the sentence 'If the engineer discovers gaps... the system triggers an automatic email notification, routing the task log back to the expert to unblock development' is slightly wordy. Consider tightening to '...the system triggers an automatic notification to the expert to unblock development'.